La Scuola d'Italia / Huson / Physics
29 May 2026

First & last name:
Grade:

3.6 Problem Set: Mechanical Advantage

Formulas (provided). $\text{MA} = \dfrac{F_{\text{out}}}{F_{\text{in}}}$ Pulley (ideal): MA = number of supporting strands Lever: $\text{MA} = \dfrac{\text{effort arm}}{\text{load arm}}$

Pressure $P = \dfrac{F}{A}$; Pascal: $\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$ Circle area $A = \pi r^2$ Chain at midpoint: $F_{\text{pull}} = 2\,T\sin\theta$

1 Pa = 1 N/m^2 1 kPa = 1000 Pa 1 PSI = 6.895 kPa $g = 9.8\ \text{m/s}^2$

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1. **Pulleys.** A warehouse worker lifts crates with three pulley setups.

   (a) State the ideal mechanical advantage of a single *fixed* pulley, and say what it is good for.

   (b) A single *movable* pulley supports a 240 N load on two strands. What effort force ideally lifts it?

   (c) A block and tackle has 4 strands supporting the load. Find the ideal effort force to lift a 600 N crate, and the length of rope you must pull to raise the crate 0.50 m.

2. **Lever.** A steel bar is used as a lever to lift a 850 N rock. The fulcrum sits 0.20 m from the rock (load arm); you push down at a point 1.20 m from the fulcrum (effort arm).

   (a) Find the ideal mechanical advantage.

   (b) Find the effort force needed to balance the rock (ideal case).

   (c) You push the long end down 0.30 m. How far does the rock rise? Comment on the trade-off.

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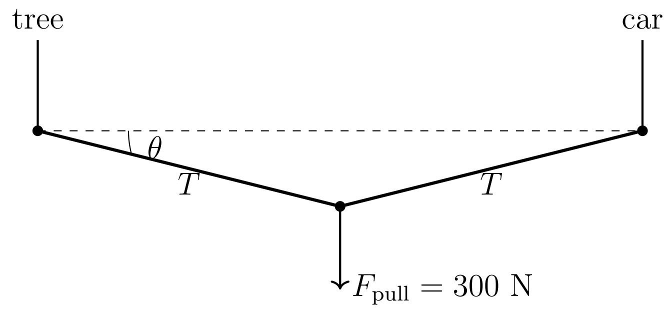
3. **Chain and stuck car.** A chain is stretched tight and straight between a tree and a car stuck in a ditch. You pull *sideways* at the midpoint with a force $F_{\text{pull}} = 300$ N, deflecting the chain by $\theta = 10°$ from the straight line. The sideways pull is balanced by the two chain tensions: $F_{\text{pull}} = 2\,T \sin\theta$.

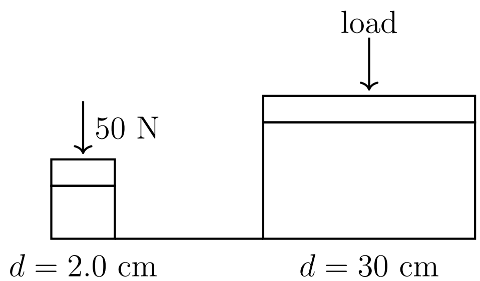


(a) Find the tension T in the chain.

(b) The mechanical advantage on the car is $\text{MA} = \dfrac{T}{F_{\text{pull}}} = \dfrac{1}{2\sin\theta}$. Evaluate it.

(c) As the chain is pulled tighter and θ gets *smaller*, what happens to the force on the car? Explain in one or two sentences, and state one practical danger.

4. **Hydraulic lift.** A hydraulic lift has a small input piston of diameter 2.0 cm and a large output piston of diameter 30 cm.



(a) Find the area of each piston in cm^2 (use $A = \pi r^2$; the radius is half the diameter).

(b) Use Pascal's principle to find the mechanical advantage $\dfrac{A_{\text{large}}}{A_{\text{small}}}$.

(c) A force of 50 N pushes the small piston. Find the output force on the large piston.

(d) The system is rated for an operating pressure of 1500 PSI. Convert this to kPa. Show one line of work using the unit factor 1 PSI = 6.895 kPa.